Minimax rate for the small-conductivity Darcy inverse problem

Abstract

Let $D \subset \mathbb{R}^d$, $d \geq 2$, be the unit ball, and let

$$\operatorname{div}(f \nabla u_f) = 1 \quad \text{in } D, \qquad u_f|_{\partial D} = 0.$$

If $f \in C^\alpha(D)$, $\alpha > 1$, and

$$\mathcal{F}(\alpha, r) = \{f : \|f - 1\|_{C^\alpha(D)} \leq r\},$$

then, for $r > 0$ small enough, the minimax L^2-risk in the white-noise model

$$Y_n = u_f + n^{-1/2}\mathbb{W}$$

satisfies, up to logarithmic factors,

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_n - f_0\|^2_{L^2(D)} \asymp_{\log} n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

Equivalently, the L^2-loss itself has rate $n^{-\alpha/(2\alpha+d+3)}$, up to logarithms.

1 Statement of the result

Write $G(f) = u_f$. The observation model is the Gaussian shift model on $L^2(D)$: for every $\phi \in L^2(D)$,

$$Y_n(\phi) = \langle G(f), \phi \rangle_{L^2(D)} + n^{-1/2}\mathbb{W}(\phi),$$

where $\mathbb{W}$ is standard $L^2(D)$-white noise. The Kullback–Leibler divergence is

$$K(P_f^{(n)}, P_g^{(n)}) = \frac{n}{2}\|G(f) - G(g)\|^2_{L^2(D)}.$$

Theorem 1 (Minimax rate)**.** *Let $d \geq 2$ and $\alpha > 1$. There exist constants $r_0, c, C, \kappa > 0$, depending only on d, α, such that for every $0 < r \leq r_0$,*

$$c\, n^{-\frac{2\alpha}{2\alpha+d+3}} \leq \inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_n - f_0\|^2_{L^2(D)} \leq C(\log n)^\kappa n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

The exponent $d + 3$ is the important point. The linearisation at $f = 1$ is

$$DG_1[h] = -\Delta_D^{-1} \operatorname{div}\Big(h\frac{x}{d}\Big),$$

whose principal symbol is

$$\frac{i(x \cdot \xi)}{d|\xi|^2}.$$

This symbol vanishes on the codimension-one characteristic set $\{(x, \xi) : x \cdot \xi = 0\}$. Thus the problem is not elliptic of order one, nor elliptic of order two; the degeneracy contributes one additional power of frequency, giving the effective variance J^{d+3}/n.

2 PDE facts and linearisation

Let

$$L_f v = \operatorname{div}(f \nabla v), \qquad v|_{\partial D} = 0.$$

Since $\|f - 1\|_{L^\infty} \le r < 1$, L_f is uniformly elliptic. By Schauder theory for divergence-form elliptic equations with C^α-coefficients, if $\alpha > 1$, then

$$f \in C^\alpha(D) \quad \Longrightarrow \quad u_f \in C^{\alpha+1}(D).$$

Moreover, $G : f \mapsto u_f$ is C^2 on small C^α-balls, and

$$DG_f[h] = v$$

is the solution of

$$L_f v = -\operatorname{div}(h \nabla u_f), \qquad v|_{\partial D} = 0.$$

Similarly,

$$D^2 G_f[h_1, h_2] = w$$

solves

$$L_f w = -\operatorname{div}(h_1 \nabla DG_f[h_2]) - \operatorname{div}(h_2 \nabla DG_f[h_1]).$$

These are standard consequences of the implicit function theorem and elliptic Schauder estimates; see Gilbarg–Trudinger [**?**, Chs. 6, 8] or Lieberman [**?**, Ch. 6].

For $f = 1$,

$$u_1(x) = \frac{|x|^2 - 1}{2d}, \qquad \nabla u_1(x) = \frac{x}{d}.$$

Hence

$$DG_1[h] = Ah, \qquad \Delta(Ah) = -\operatorname{div}\Big(h\frac{x}{d}\Big) = -\left(h + \frac{x \cdot \nabla h}{d}\right).$$

3 The microlocal trace estimate

Let $(\mathcal{V}_J)_{J \ge 2}$ be a standard smooth multiresolution sequence on D, for instance obtained by restricting compactly supported wavelets to the ball. We use only the following properties:

$$\dim \mathcal{V}_J \asymp J^d, \qquad \inf_{v \in \mathcal{V}_J} \|g - v\|_{L^2(D)} \lesssim J^{-\alpha} \|g\|_{C^\alpha(D)}.$$

Such spaces are classical; see Triebel [**?**].

Proposition 2 (Fisher trace bound)**.** *For $r > 0$ small enough, uniformly in $f \in \mathcal{F}(\alpha, r)$, the restriction*

$$B_{f,J} = DG_f|_{\mathcal{V}_J} : \mathcal{V}_J \to L^2(D)$$

is injective and satisfies

$$\operatorname{tr}\big[(B_{f,J}^* B_{f,J})^{-1}\big] \lesssim (\log J)^{\kappa_0} J^{d+3}.$$

The logarithm can be removed for spectral cutoffs adapted to the operator, but is immaterial for the minimax rate.

Proof sketch with standard microlocal input. Write

$$DG_f = -L_f^{-1} T_f, \qquad T_f h = \operatorname{div}(h \nabla u_f).$$

The operator L_f^{-1} is elliptic of order -2, while T_f is a first-order transport operator with full symbol

$$\tau_f(x, \xi) = i \nabla u_f(x) \cdot \xi + \Delta u_f(x).$$

For r small, Schauder estimates imply

$$\|\nabla u_f - x/d\|_{C^1(D)} + \|\Delta u_f - 1\|_{L^\infty(D)} \le Cr.$$

Choosing r sufficiently small gives

$$\Delta u_f(x) \ge c_0 > 0, \qquad D(\nabla u_f)(x) \ge c_0 I$$

in the sense that $D(\nabla u_f)$ is uniformly close to I/d.

The normal operator $B_f^* B_f$ has principal symbol

$$p_f(x, \xi) \asymp \frac{|\nabla u_f(x) \cdot \xi|^2 + (\Delta u_f(x))^2}{|\xi|^4}.$$

Thus the inverse Fisher trace over frequencies $|\xi| \lesssim J$ is controlled by the phase-space integral

$$\int_D \int_{|\xi| \le J} \frac{|\xi|^4}{1 + |\nabla u_f(x) \cdot \xi|^2} \, d\xi \, dx.$$

Let $\omega = \xi / |\xi|$. Since

$$\nabla_x (\nabla u_f(x) \cdot \omega) = (D^2 u_f(x)) \omega$$

has norm bounded below uniformly in x, ω, the coarea formula gives

$$|\{x \in D : |\nabla u_f(x) \cdot \omega| \le t\}| \lesssim t, \qquad 0 < t < 1.$$

Consequently, for $\rho = |\xi| \ge 1$,

$$\int_D \frac{dx}{1 + \rho^2 |\nabla u_f(x) \cdot \omega|^2} \lesssim \rho^{-1}.$$

Therefore

$$\int_D \int_{|\xi| \le J} \frac{|\xi|^4}{1 + |\nabla u_f(x) \cdot \xi|^2} \, d\xi \, dx \lesssim \int_1^J \rho^{d-1} \rho^4 \rho^{-1} \, d\rho \lesssim J^{d+3}.$$

The passage from this phase-space bound to the matrix trace $\operatorname{tr}[(B_{f,J}^* B_{f,J})^{-1}]$ is the standard semiclassical trace estimate for elliptic pseudodifferential operators with parameter; see Birman–Solomyak [**?**] and Hörmander [**?**, Vol. III, Ch. 18]. The logarithmic factor accounts for using a dyadic multiresolution space instead of an exact spectral cutoff. □

4 Upper bound

Let $J = J_n$ be chosen later and define the sieve

$$\mathcal{F}_J = \mathcal{F}(\alpha, r) \cap (1 + \mathcal{V}_J).$$

Let $\hat{f}_J$ be any measurable minimiser over $\mathcal{F}_J$ of the Gaussian least-squares contrast

$$\gamma_n(f) = \|G(f)\|^2_{L^2(D)} - 2Y_n(G(f)).$$

Existence follows by compactness of $\mathcal{F}_J$.

A standard finite-dimensional likelihood theorem for nonlinear Gaussian shift models gives

$$\sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_J - f_0\|^2_{L^2(D)} \lesssim J^{-2\alpha} + \frac{(\log J)^{\kappa_0} J^{d+3}}{n}.$$

Indeed, the first term is the approximation error

$$\inf_{g \in \mathcal{F}_J} \|g - f_0\|^2_{L^2(D)} \lesssim J^{-2\alpha},$$

and the second term is the trace of the inverse Fisher information divided by n. The required local quadratic expansion of the likelihood follows from the C^2-differentiability of G and the tangential-cone estimate obtained from the equation for $D^2 G_f$. This is precisely the sieve likelihood argument used for PDE-constrained regression; see Ibragimov–Has'minskii [**?**, Ch. I], van der Vaart [**?**, Ch. 7], and Nickl–van de Geer–Wang [**?**] for the nonlinear elliptic PDE setting.

Balancing

$$J^{-2\alpha} \asymp \frac{J^{d+3}}{n}$$

gives

$$J_n \asymp \left(\frac{n}{\log n}\right)^{1/(2\alpha+d+3)}.$$

Therefore

$$\sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_{J_n} - f_0\|^2_{L^2(D)} \lesssim (\log n)^{\kappa} n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

5 Lower bound

We prove a matching Assouad lower bound.

Fix a smooth radial cut-off $\eta \in C_c^\infty((1/2, 3/4))$. On S^{d-1}, choose a needlet family

$$\{\psi_{J,k} : 1 \le k \le M_J\}, \qquad M_J \asymp J^{d-1},$$

with angular frequencies in $[J/2, 2J]$, uniformly bounded overlap, and normalisation

$$\|\psi_{J,k}\|_{L^\infty(S^{d-1})} \lesssim 1, \qquad \|\psi_{J,k}\|^2_{L^2(S^{d-1})} \asymp J^{-(d-1)}.$$

Such needlets exist on compact homogeneous manifolds; see Narcowich–Petrushev–Ward [**?**]. Define

$$\phi_{J,k}(x) = \eta(|x|)\psi_{J,k}\left(\frac{x}{|x|}\right).$$

Then

$$\|\phi_{J,k}\|^2_{L^2(D)} \asymp J^{-(d-1)}, \qquad \|\phi_{J,k}\|_{C^\alpha(D)} \lesssim J^\alpha.$$

For $\theta = (\theta_1, \dots, \theta_{M_J}) \in \{-1, 1\}^{M_J}$, set

$$f_\theta = 1 + \gamma_J \sum_{k=1}^{M_J} \theta_k \phi_{J,k}, \qquad \gamma_J = c_r J^{-\alpha},$$

with $c_r > 0$ sufficiently small. The Besov characterisation of needlet coefficients gives

$$\|f_\theta - 1\|_{C^\alpha(D)} \le r,$$

so $f_\theta \in \mathcal{F}(\alpha, r)$.

If θ, θ' differ in exactly one coordinate, then

$$\|f_\theta - f_{\theta'}\|^2_{L^2(D)} \asymp \gamma_J^2 J^{-(d-1)}.$$

More generally,

$$\|f_\theta - f_{\theta'}\|^2_{L^2(D)} \gtrsim \gamma_J^2 J^{-(d-1)} d_H(\theta, \theta'),$$

where d_H denotes Hamming distance.

It remains to control the KL divergence between neighbouring hypotheses. Let $\theta^{(k)}$ be obtained from θ by flipping the k-th coordinate. By the mean-value formula,

$$G(f_\theta) - G(f_{\theta^{(k)}}) = 2\gamma_J \int_0^1 DG_{f_{\theta^{(k)}} + t(f_\theta - f_{\theta^{(k)}})}[\phi_{J,k}]\, dt.$$

Along this segment the conductivities converge to 1 in C^1, because

$$\gamma_J J = c_r J^{1-\alpha} \to 0 \qquad (\alpha > 1).$$

For $f = 1$, the action of DG_1 on $\phi_{J,k}$ is order J^{-2}: indeed,

$$\Delta DG_1[\phi_{J,k}] = -\left(\phi_{J,k} + \frac{x \cdot \nabla \phi_{J,k}}{d}\right),$$

and the right-hand side has angular frequencies $\asymp J$ but only bounded radial frequency. The inverse Dirichlet Laplacian therefore gains two angular derivatives:

$$\|DG_1[\phi_{J,k}]\|_{L^2(D)} \lesssim J^{-2} \|\phi_{J,k}\|_{L^2(D)}.$$

The same bound holds uniformly along the above segment by perturbation, since the segment is $o(1)$-close to 1 in C^1. Thus

$$\|G(f_\theta) - G(f_{\theta^{(k)}})\|^2_{L^2(D)} \lesssim \gamma_J^2 J^{-4} J^{-(d-1)}.$$

Consequently,

$$K(P^{(n)}_{f_\theta}, P^{(n)}_{f_{\theta^{(k)}}}) \lesssim n\gamma_J^2 J^{-4} J^{-(d-1)} = n J^{-2\alpha - d - 3}.$$

Choose

$$J \asymp n^{1/(2\alpha + d + 3)}$$

with a sufficiently large implicit constant, so that the neighbouring KL divergences are bounded by a fixed small number. Assouad's lemma [**?**, Thm. 2.12] yields

$$\inf_{\hat{f}_n} \sup_{\theta\in\{-1,1\}^{M_J}} \mathbb{E}_\theta \|\hat{f}_n - f_\theta\|^2_{L^2(D)} \gtrsim M_J \gamma_J^2 J^{-(d-1)} \asymp \gamma_J^2 \asymp J^{-2\alpha}.$$

With the above choice of J,

$$J^{-2\alpha} \asymp n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

This proves the lower bound.

Remark 3. *The lower bound uses perturbations that oscillate tangentially to the radial vector field x/d. For such perturbations the first-order part $x\cdot\nabla$ is essentially inactive, so the linearised forward map smooths by two derivatives. However, only $d-1$ tangential directions are available. Thus the hard alternatives have dimension $d-1$ and ill-posedness 2, giving denominator*

$$2\alpha + 2\cdot 2 + (d-1) = 2\alpha + d + 3.$$

This matches the upper trace calculation.

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